

Sanjana Venkatesh

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EDUCATION

B.E. Computer Science and Engineering | RNS Institute of Technology, Bengaluru | CGPA: 8.38/10 2023-2027

EXPERIENCE

IoT Intern | Samsung Innovation Campus

Sep 2025 - Present

- Selected among **800+ applicants** for Samsung's competitive Innovation Campus program, completing intensive training in Internet of Things, embedded systems, and sensor integration.
- Led a team of 4** in designing and building an end-to-end IoT project, coordinating task delegation, sprint planning, and technical reviews to ensure timely and quality delivery.
- Drove team decisions on architecture and technology stack, facilitating discussions across hardware, firmware, and web layers to align the group on a cohesive system design.
- Built a real-time smart monitoring system using Raspberry Pi Pico (RP2040) and MicroPython, leveraging SPI protocol and GPIO for seamless sensor data acquisition and actuation.
- Designed and deployed a web interface to monitor and control IoT device states in real time, gaining hands-on experience in full-stack IoT development from hardware to cloud.
- Mentored team members on embedded systems concepts and debugging practices, fostering a collaborative learning environment throughout the program.

Software Developer Intern | IEEE TechXcelerate

Jun 2025 - Oct 2025

- Engineered a physics problem simulator using Node.js and Express.js, enabling real-time 2D rigid body dynamics simulations with full-stack development approach.
- Implemented RAG (Retrieval-Augmented Generation) pipeline using Gemini-embedding-001 to parse complex user physics statements and map them to simulation using cosine similarity parameters achieving high accuracy.
- Designed and managed PostgreSQL schema to store and retrieve simulation states and user-generated problem templates.
- Integrated Matter.js and P5.js with RESTful API to render dynamic, parameter-driven visual simulations based on AI-parsed inputs.
- Architected and deployed a library of 39+ modular physics simulations (Kinematics, Electromagnetism, Dynamics) using P5.js and Matter.js, ensuring real-time rendering and physical accuracy for complex user-defined parameters.

PROJECTS

TimeTrace AI: VS Code Debugging Timeline Extension | TypeScript, VS Code API

Apr 2026 – Apr 2026

- Engineered a VS Code extension that builds save-by-save checkpoints and detects risky changes, surfacing probable root causes in a clean timeline sidebar interface.
- Designed an AI-style analysis engine performing feature extraction, finding detection, risk classification, and root-cause ranking across related modules on every file save event.
- Implemented dependency impact mapping and runtime signal correlation to strengthen root-cause confidence and track incident lifecycle from detection to resolution.
- Architected the webview sidebar (HTML/CSS/JS) with interactive scenario support and a scripted demo workspace for reproducible evaluations and live walkthroughs.

SimuPhysics: AI-Powered Physics Simulation Platform | Full Stack, AI

Jun 2025 - Oct 2025

- Full-stack development of physics problem simulator using HTML, CSS, JavaScript, Matter.js, Google Gemini, REST API, Node.js, and Express.js with real-time 2D rigid body dynamics simulations.
- Implemented AI and RAG Architecture using Gemini-embedding-001 for parsing complex user physics statements and mapping them to simulation parameters with cosine similarity achieving high accuracy.
- Designed PostgreSQL database schema for simulation states and user-generated templates with frontend integration using Matter.js and P5.js for dynamic visual rendering.

BleepBloop: Real-Time Full-Stack Chat Application | WebSockets, Java

Feb 2025 – Mar 2026

- Engineered a full-stack instant messaging platform using Spring Boot 3 and React 19, enabling bi-directional, low-latency communication via WebSockets (STOMP/SockJS).
- Implemented Spring Security for user authentication and authorization, securing REST endpoints and managing WebSocket handshake interceptors.
- Architected a relational schema in PostgreSQL using Spring Data JPA to manage persistent user data, online presence tracking, and cryptographic public key storage.
- Developed a responsive UI with TailwindCSS and Vite, utilizing StompJS for client-side message brokering and private routing to specific user queues.
- Containerized the database environment using Docker Compose to ensure environment parity and streamlined local development workflows.

LEADERSHIP & ACTIVITIES

Core Team Member, BigO Club (Competitive Coding Club): Hosted several events including Intro to Git and coding competitions for fellow peers. (Oct 2024 - Present)

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python, C, C++, Java

Backend Development: Node.js, Express.js, Spring Boot, FastAPI, Django, RESTful APIs

Frontend Development: React.js, P5.js, Matter.js, HTML5, CSS3

Databases: MySQL, PostgreSQL, Gemini Embeddings

Tools & Version Control: Git (CLI & GUI), GitHub, Docker

Cloud & Platforms: Linux (Ubuntu terminal), Windows, AWS (EC2/S3)

AI/ML: Generative AI (Gemini 2.5 Pro), RAG Architecture, NLP, Computer Vision

Embedded Systems: Raspberry Pi Pico (RP2040), MicroPython, SPI Protocol, GPIO

CERTIFICATIONS & ACHIEVEMENTS

- Samsung Innovation Campus - IoT:** Selected among **800+ applicants**. Completed intensive training in Internet of Things, embedded systems, sensor integration, and IoT application development.
- NPTEL Certifications:** Earned two certifications in core computer science and engineering domains, demonstrating strong foundational knowledge.
- Competitive Programming:** Solved **150+ problems** on LeetCode, demonstrating strong proficiency in data structures and algorithms.
- Minor Degree in VLSI:** Currently pursuing minor degree in VLSI (Very Large Scale Integration), expanding expertise in semiconductor design and digital systems.